Posterior Calibration Report: Ballistics and Kinematics Benchmarks

This report collects the calibration diagnostics for every model on the two benchmarks. Each benchmark has five plots: coverage curves, rank histograms, signed coverage error, spread-ratio distribution, and the spread-vs-calibration comparison. Each plot is followed by a short explanation of how to read it and what it shows.

How to read the diagnostics

Marginal coverage checks, one dimension at a time, whether central credible intervals at nominal level α contain the true value a fraction α of the time. **Joint (HPD-like) coverage** does the same for highest-density regions over all dimensions together, so it can catch problems in the dependence structure that the marginals hide.

Signed coverage error is empirical minus nominal coverage, averaged over levels. Positive means the model's intervals are too wide (**under-confident**); negative means they are too narrow (**over-confident**). A model counts as **calibrated** if the error is within ±0.03.

Rank histograms show the rank of the true parameter among the posterior samples. Flat means calibrated, a U-shape means over-confident (truth often falls in the tails), and a hump means under-confident (truth is concentrated in the middle).

Spread ratio compares the model's posterior spread with the ground-truth posterior spread, as $\log_2$(model / ground truth). Zero is correct, positive is too wide, negative is too narrow. The ground-truth check is independent of the coverage check, so the two can confirm or contradict each other.

Table columns

marg / joint: signed coverage error (marginal, joint). **verdicts**: calibrated if |error| ≤ 0.03. **tails**: rank-tail mass; about 0.10 is expected when calibrated, lower goes with a hump (under-confident) and higher with a U-shape (over-confident). **spread/GT**: median ratio of model spread to ground-truth spread. **GT says**: verdict from the spread ratio. **agree**: whether the GT verdict matches the marginal coverage verdict. **nonfin**: share of non-finite samples (0% for every model).

Summary tables

Ballistics (20 models)

model	marg	marginal verdict	joint	joint verdict	tails	spread/GT	GT says	agree	nonfin
bal_K16_s0	-0.002	calibrated	-0.008	calibrated	0.100	1.00	calibrated	yes	0%
bal_cfmpe_s0_midpointx16	-0.002	calibrated	-0.005	calibrated	0.098	1.00	calibrated	yes	0%
bal_cinn_s0	+0.029	calibrated	-0.006	calibrated	0.069	1.12	under-confident	no	0%
bal_cinn_s1	+0.000	calibrated	+0.006	calibrated	0.098	1.00	calibrated	yes	0%
bal_cinn_s2	+0.013	calibrated	-0.006	calibrated	0.099	1.04	calibrated	yes	0%
bal_cvae_s0	-0.002	calibrated	-0.006	calibrated	0.102	1.00	calibrated	yes	0%
bal_fff_s0	-0.003	calibrated	-0.004	calibrated	0.099	1.00	calibrated	yes	0%
bal_fif_s0	+0.000	calibrated	+0.005	calibrated	0.099	1.00	calibrated	yes	0%
bal_fmpe_s0_midpointx16	-0.001	calibrated	-0.003	calibrated	0.099	1.00	calibrated	yes	0%
bal_ilnn_ml_s0	+0.000	calibrated	-0.008	calibrated	0.097	1.00	calibrated	yes	0%
bal_ilnn_mmd_s0	-0.001	calibrated	-0.002	calibrated	0.100	1.00	calibrated	yes	0%
bal_ilnn_s0	-0.002	calibrated	-0.003	calibrated	0.100	1.00	calibrated	yes	0%
bal_inn_l2mmd_s0	-0.001	calibrated	-0.002	calibrated	0.097	1.00	calibrated	yes	0%
bal_inn_l2mmd_s1	-0.000	calibrated	-0.001	calibrated	0.098	1.00	calibrated	yes	0%
bal_inn_l2mmd_s2	-0.002	calibrated	-0.008	calibrated	0.100	1.05	calibrated	yes	0%
bal_inn_ml_s0	+0.057	under-confident	+0.044	under-confident	0.054	1.29	under-confident	yes	0%
bal_inn_ml_s1	+0.095	under-confident	+0.100	under-confident	0.037	1.39	under-confident	yes	0%
bal_inn_ml_s2	+0.067	under-confident	+0.076	under-confident	0.051	1.34	under-confident	yes	0%
bal_invertible_s0	-0.002	calibrated	+0.001	calibrated	0.100	1.00	calibrated	yes	0%
bal_standard_s0	+0.001	calibrated	-0.001	calibrated	0.098	1.00	calibrated	yes	0%

Kinematics (20 models)

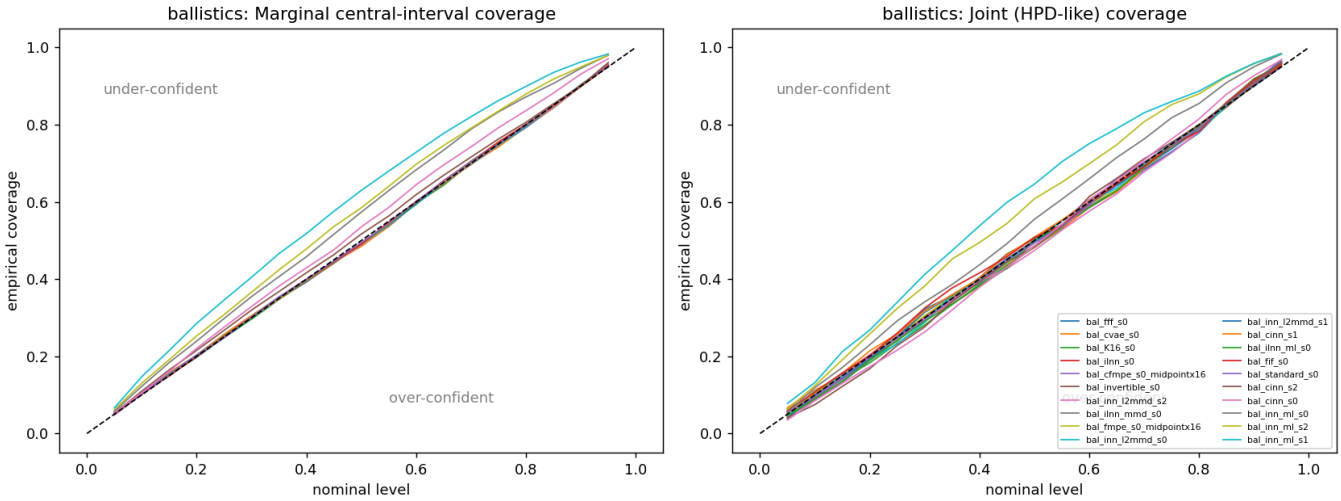
model	marg	marginal verdict	joint	joint verdict	tails	spread/GT	GT says	agree	nonfin
kin_K16_s0	-0.004	calibrated	-0.036	over-confident	0.103	1.01	calibrated	yes	0%
kin_cfmpe_s0_midpointx16	-0.006	calibrated	-0.006	calibrated	0.103	1.00	calibrated	yes	0%
kin_cinn_s0	-0.002	calibrated	-0.008	calibrated	0.101	1.01	calibrated	yes	0%
kin_cinn_s1	-0.005	calibrated	-0.010	calibrated	0.101	1.00	calibrated	yes	0%
kin_cinn_s2	-0.001	calibrated	-0.011	calibrated	0.101	1.01	calibrated	yes	0%
kin_cvae_s0	+0.005	calibrated	+0.005	calibrated	0.086	1.06	calibrated	yes	0%
kin_fff_s0	-0.004	calibrated	+0.036	under-confident	0.099	1.00	calibrated	yes	0%
kin_fif_s0	-0.005	calibrated	+0.031	under-confident	0.093	1.01	calibrated	yes	0%
kin_fmpe_s0_midpointx16	-0.006	calibrated	-0.006	calibrated	0.104	1.00	calibrated	yes	0%
kin_ilnn_ml_s0	+0.003	calibrated	-0.044	over-confident	0.100	1.01	calibrated	yes	0%
kin_ilnn_mmd_s0	-0.022	calibrated	-0.115	over-confident	0.132	1.00	calibrated	yes	0%
kin_inn_l2mmd_s0	+0.013	calibrated	-0.046	over-confident	0.080	1.28	under-confident	no	0%
kin_inn_l2mmd_s1	+0.002	calibrated	-0.051	over-confident	0.094	1.05	calibrated	yes	0%
kin_inn_l2mmd_s2	+0.002	calibrated	-0.064	over-confident	0.092	1.05	calibrated	yes	0%
kin_inn_ml_s0	+0.001	calibrated	-0.046	over-confident	0.099	1.00	calibrated	yes	0%
kin_inn_ml_s1	+0.001	calibrated	-0.045	over-confident	0.097	1.00	calibrated	yes	0%
kin_inn_ml_s2	+0.003	calibrated	-0.044	over-confident	0.097	1.00	calibrated	yes	0%
kin_invertible_s0	-0.008	calibrated	-0.074	over-confident	0.107	1.03	calibrated	yes	0%
kin_mfff_wrapped_b400_s0	-0.004	calibrated	+0.034	under-confident	0.099	1.00	calibrated	yes	0%
kin_standard_s0	-0.003	calibrated	-0.024	calibrated	0.099	1.02	calibrated	yes	0%

Colour key: green = calibrated, red = over-confident, blue = under-confident. A red **no** in the agree column marks a model where the ground-truth spread check and the marginal coverage check disagree.

Ballistics

Overall: 17 of 20 models are calibrated on both marginal and joint coverage, and their spread matches the ground truth (ratio 1.00 to 1.05). The three **bal_inn_ml** seeds (s0, s1, s2) are clearly under-confident, with posteriors 29% to 39% wider than the truth. **bal_cinn_s0** is a borderline case: its marginal error (+0.029) is just inside the threshold, but its spread is 12% too wide, so the two checks disagree.

Ballistics 1: Marginal and joint coverage curves

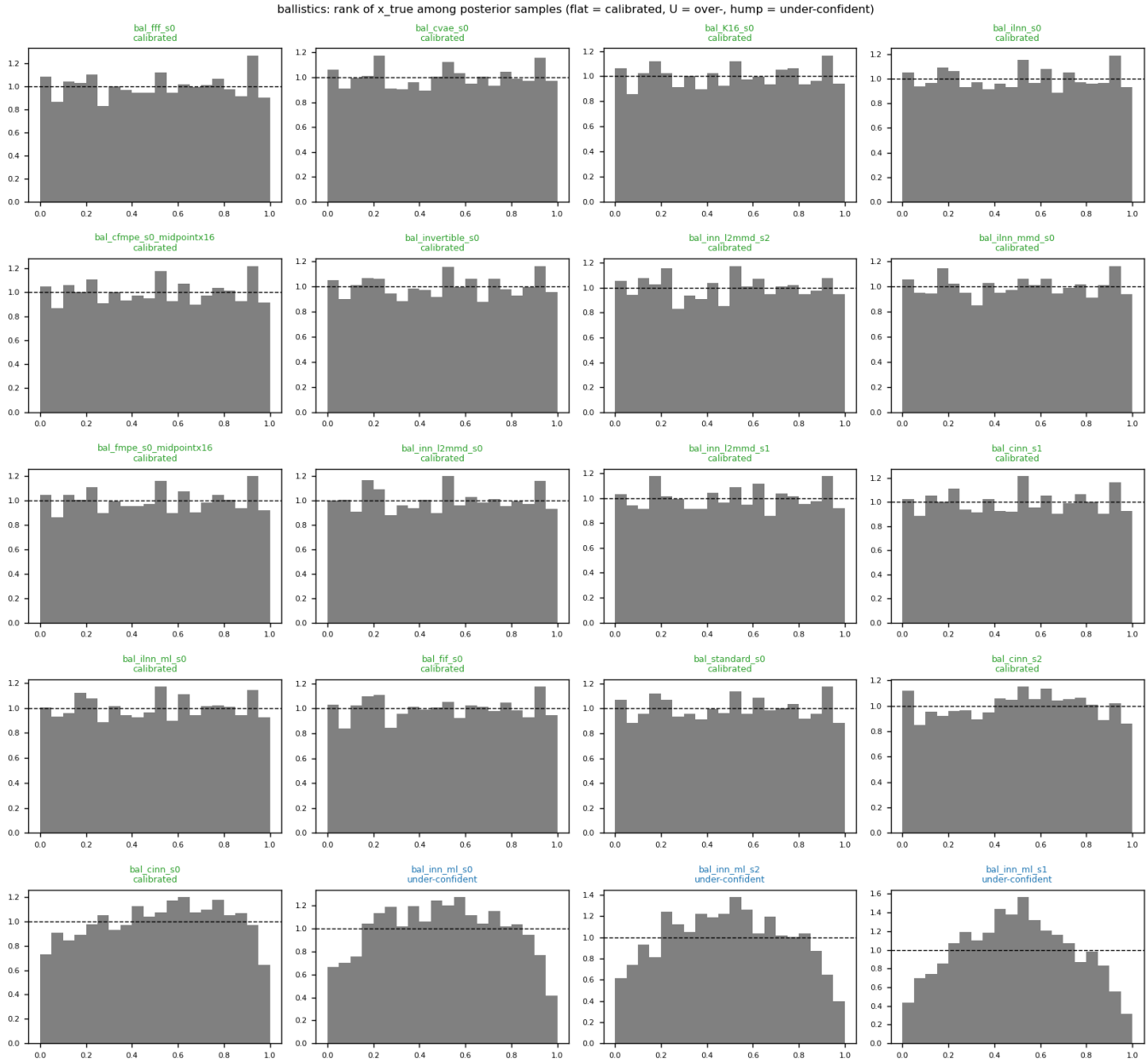


Each line shows empirical coverage against nominal level. The dashed diagonal is perfect calibration. Lines above it are under-confident; lines below it are over-confident.

Left (marginal): most models lie on the diagonal. A group of lines sits clearly above it, with the largest gap at mid-range levels (about 0.3 to 0.6). This is the under-confident group, mainly the **inn_ml** models, with **bal_cinn_s0** and **bal_inn_l2mmd_s2** slightly above the diagonal.

Right (joint): the picture is similar but noisier. The same models are above the diagonal, and the others scatter closely around it. No ballistics model falls clearly below the diagonal, so none is over-confident.

Ballistics 2: Rank histograms

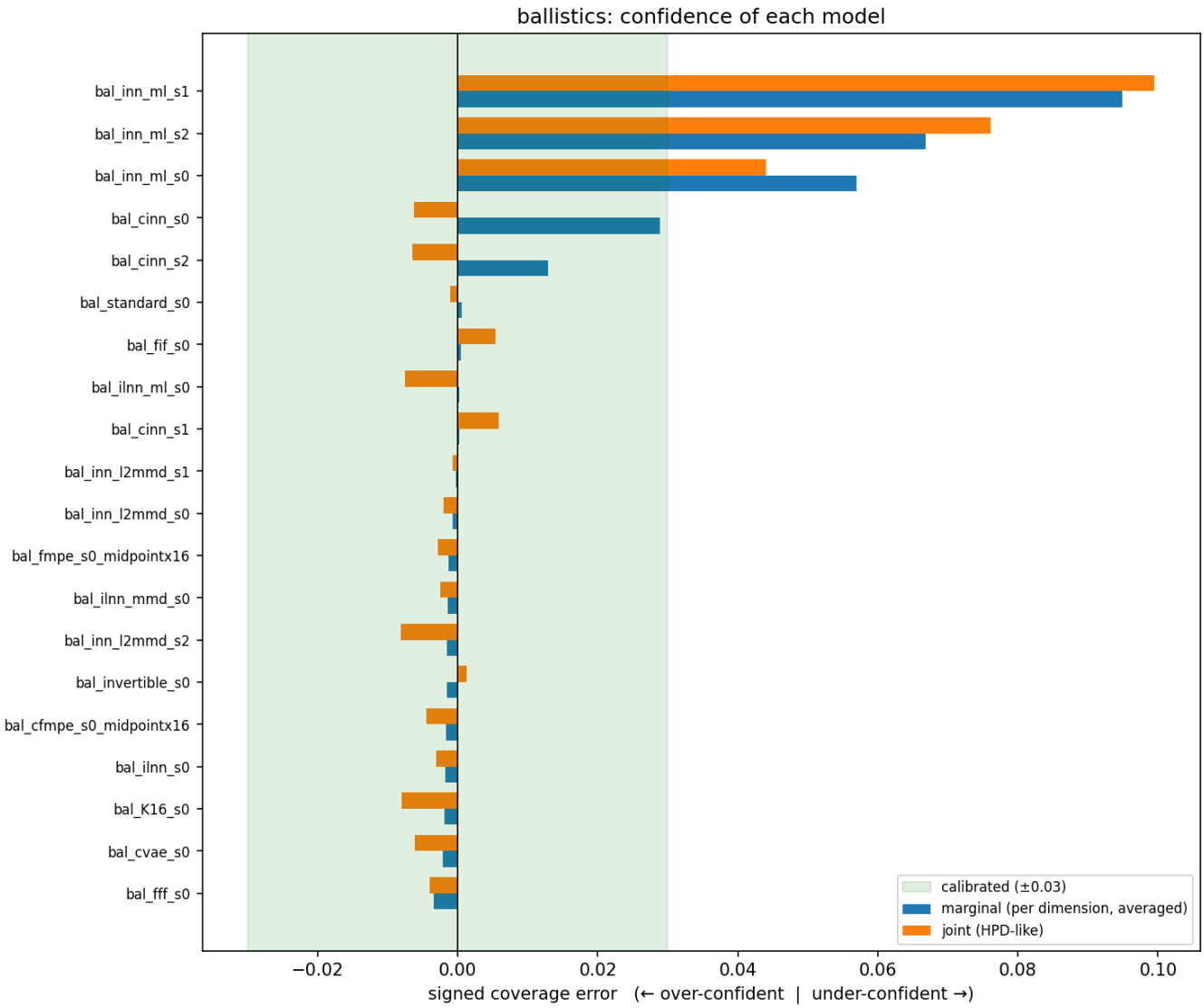


Each panel shows where the true parameter ranks among the posterior samples. The dashed line marks the uniform density of 1.0.

Seventeen panels are flat apart from sampling noise and are labelled **calibrated** (green). The three **bal_inn_ml** panels (s0, s1, s2) have a clear hump: too few truths fall in the tails and too many in the middle, which is the signature of an under-confident posterior. The effect is strongest for s1 and s2, where the outermost bins drop to about 0.3 to 0.4 of the expected height.

bal_cinn_s0 is labelled calibrated but shows a mild hump with slightly depleted tails, consistent with its small excess width.

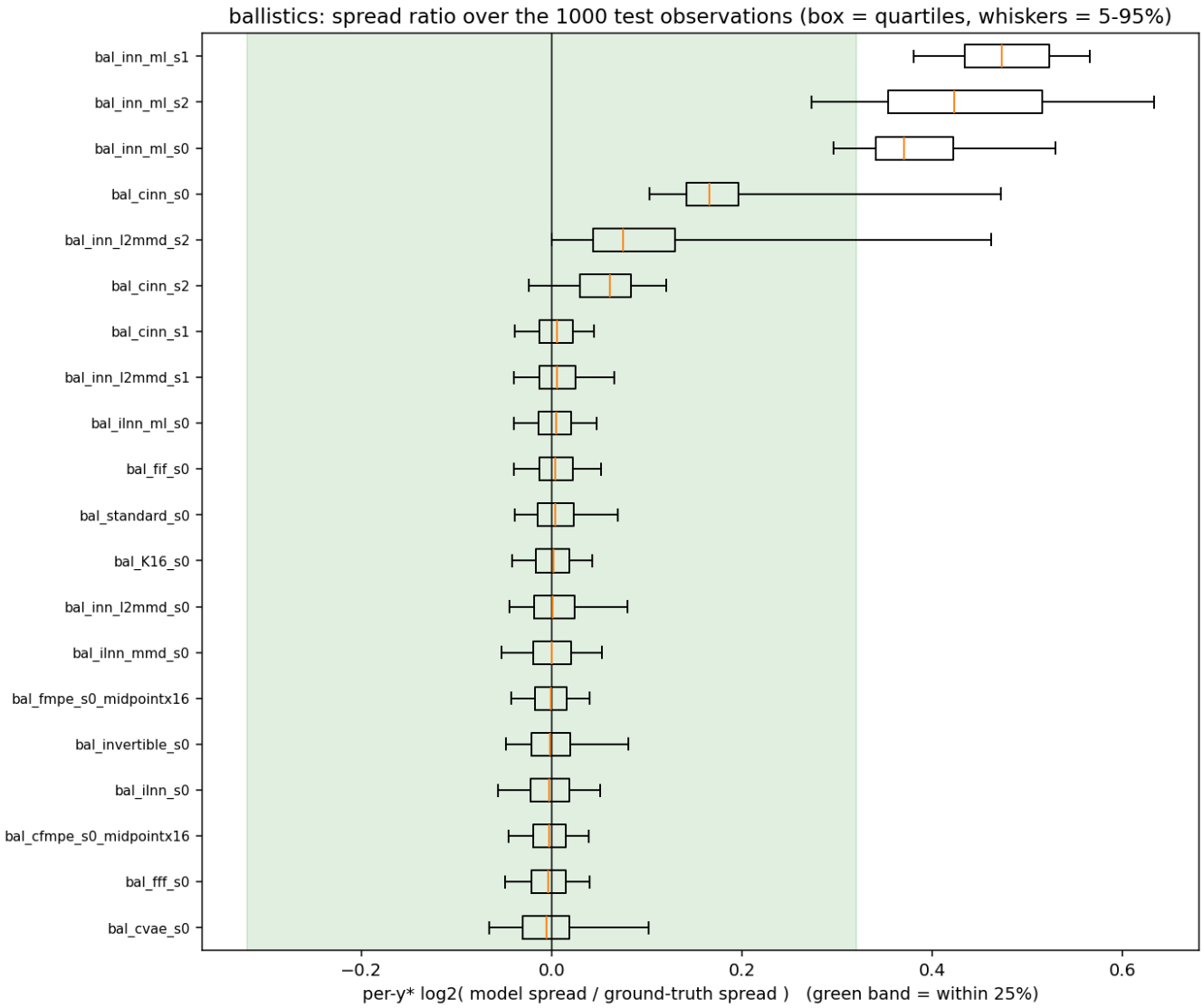
Ballistics 3: Signed coverage error per model



Bars show signed coverage error for marginal (blue) and joint (orange) coverage. The green band is the ±0.03 calibration tolerance. Bars to the right are under-confident; bars to the left are over-confident.

The three **bal_inn_ml** seeds are far outside the band on the under-confident side (joint error up to about +0.10 for s1). **bal_cinn_s0** has a marginal error of about +0.029, right at the edge of the band, while its joint error is slightly negative. All remaining models have errors within about ±0.01, well inside the band.

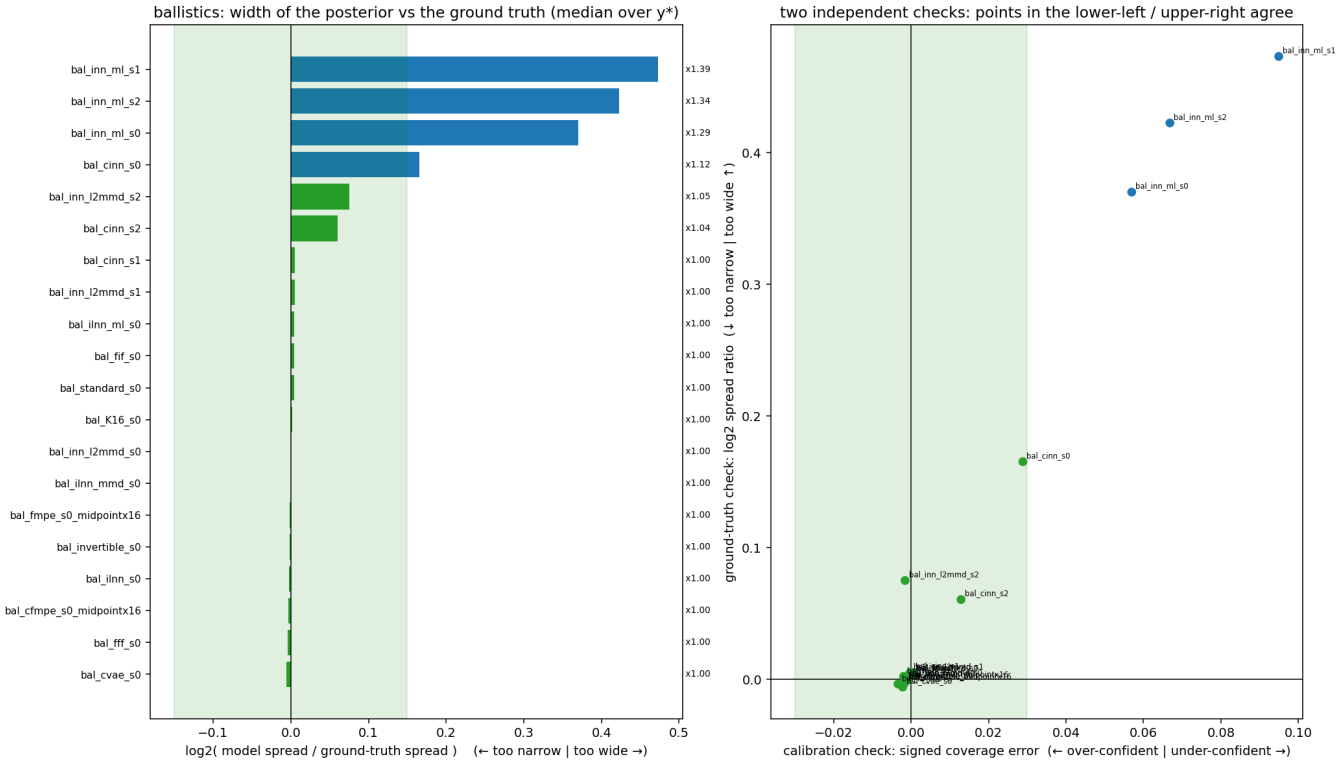
Ballistics 4: Spread ratio over the 1000 test observations



For each model, the box shows the quartiles of $\log_2$(model spread / ground-truth spread) across test observations, and the whiskers show the 5th to 95th percentiles. The green band marks ±25%.

The **inn_ml** models sit entirely to the right of the band's centre, with medians of about 0.37 to 0.47 (a factor of about 1.3 to 1.4). This means the posterior is wider than the truth for almost every test observation, not just for a few. **bal_cinn_s0** is moderately wide (median about 0.17). **bal_inn_l2mmd_s2** has a small median but a long right tail. All other models are centred on zero with tight boxes.

Ballistics 5: Posterior width vs ground truth, and agreement with coverage



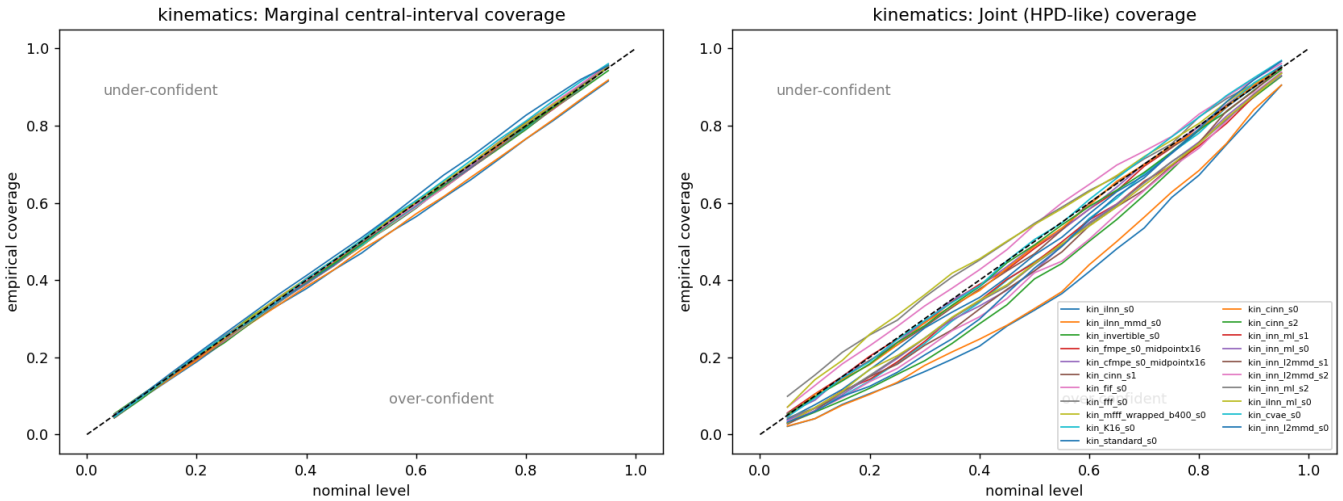
Left: median $\log_2$ spread ratio per model. Blue bars fall outside the ±0.15 band (too wide); green bars are inside. The text on the right gives the spread ratio (for example ×1.39).

Right: each model is a point, with signed coverage error on the x-axis and spread ratio on the y-axis. Two independent checks agree when a point is in the lower-left or upper-right. The three **inn_ml** models are in the upper right, so both checks agree they are under-confident. **bal_cinn_s0** is high on the spread axis but still within the coverage tolerance, which is why the table marks it as *no* agreement. The remaining models cluster at the origin.

Kinematics

Overall: every model passes the marginal coverage check (all errors within ±0.03), so the marginal view alone would call everything calibrated. The **joint** view is more revealing. Ten models are over-confident jointly (K16, ilnn_ml, ilnn_mmd, inn_l2mmd s0/s1/s2, inn_ml s0/s1/s2, invertible), and three are slightly under-confident (fff, fif, mfff_wrapped_b400). The worst joint error is **kin_ilnn_mmd_s0** at -0.115. Spread matches the ground truth for nearly all models; **kin_inn_l2mmd_s0** is the exception, with a very heavy tail of overly wide posteriors.

Kinematics 1: Marginal and joint coverage curves



Left (marginal): all models lie almost exactly on the diagonal. Two lines sit slightly below it at higher levels (about 0.92 at nominal 0.95), but the deviations are small.

Right (joint): the lines fan out. Many fall clearly below the diagonal, which indicates over-confidence, with the largest gaps at mid-range levels. A few lines sit on or slightly above the diagonal at some levels (the under-confident group). This confirms that the models capture each dimension well but not always the dependence between dimensions.

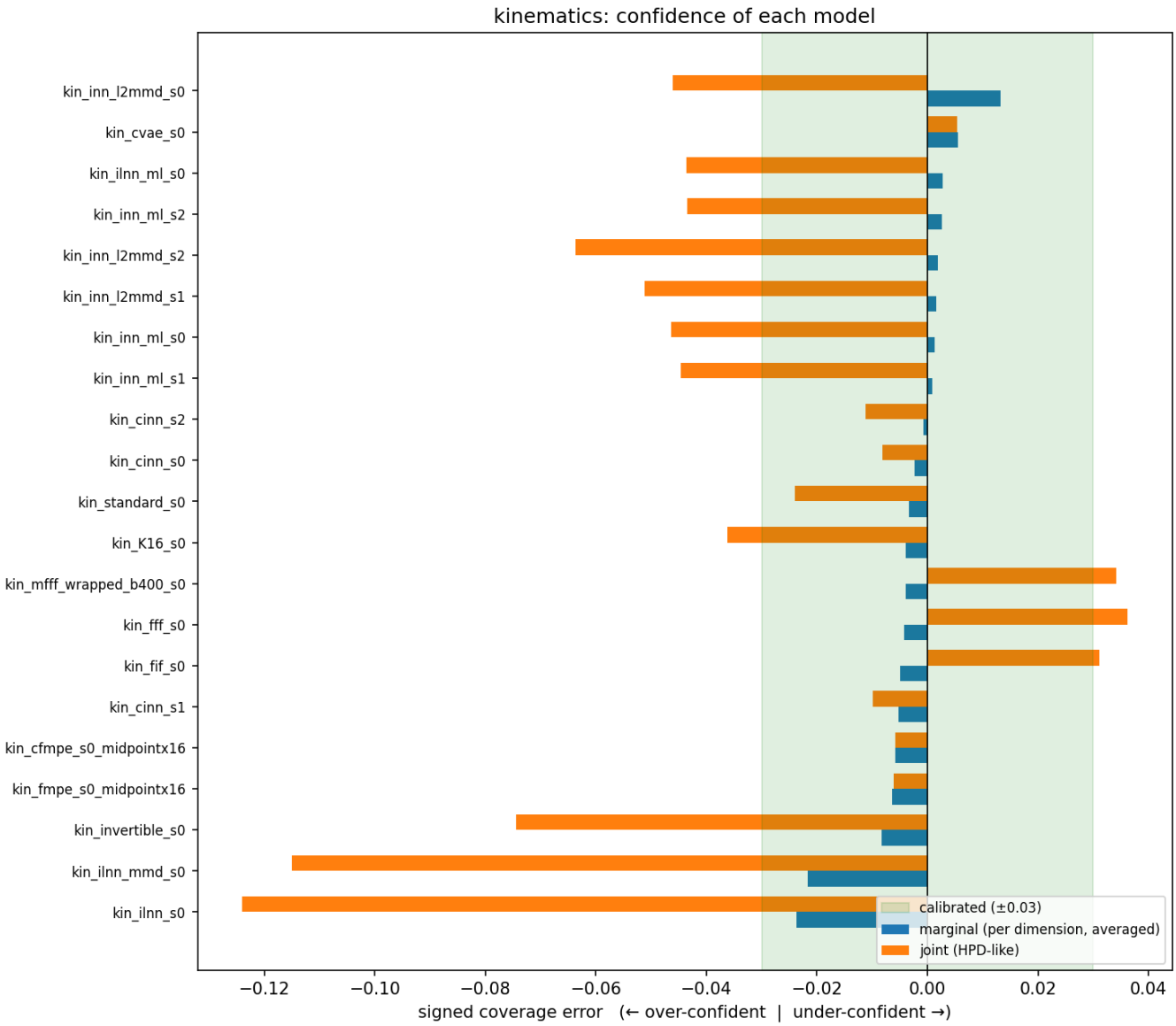
Kinematics 2: Rank histograms



Rank histograms for the marginal rank of the true value. All panels are labelled **calibrated** and look close to flat. A few show mild U-shaped edges, for example the ilnn_mmd panel, which has slightly raised first and last bins (about 1.3). kin_inn_l2mmd_s0 shows a slight depletion in the last bin.

Together with the marginal coverage curves, this shows that per-dimension calibration is good for every model. The over-confidence seen in the joint coverage is therefore a property of the joint posterior, which these marginal histograms cannot reveal.

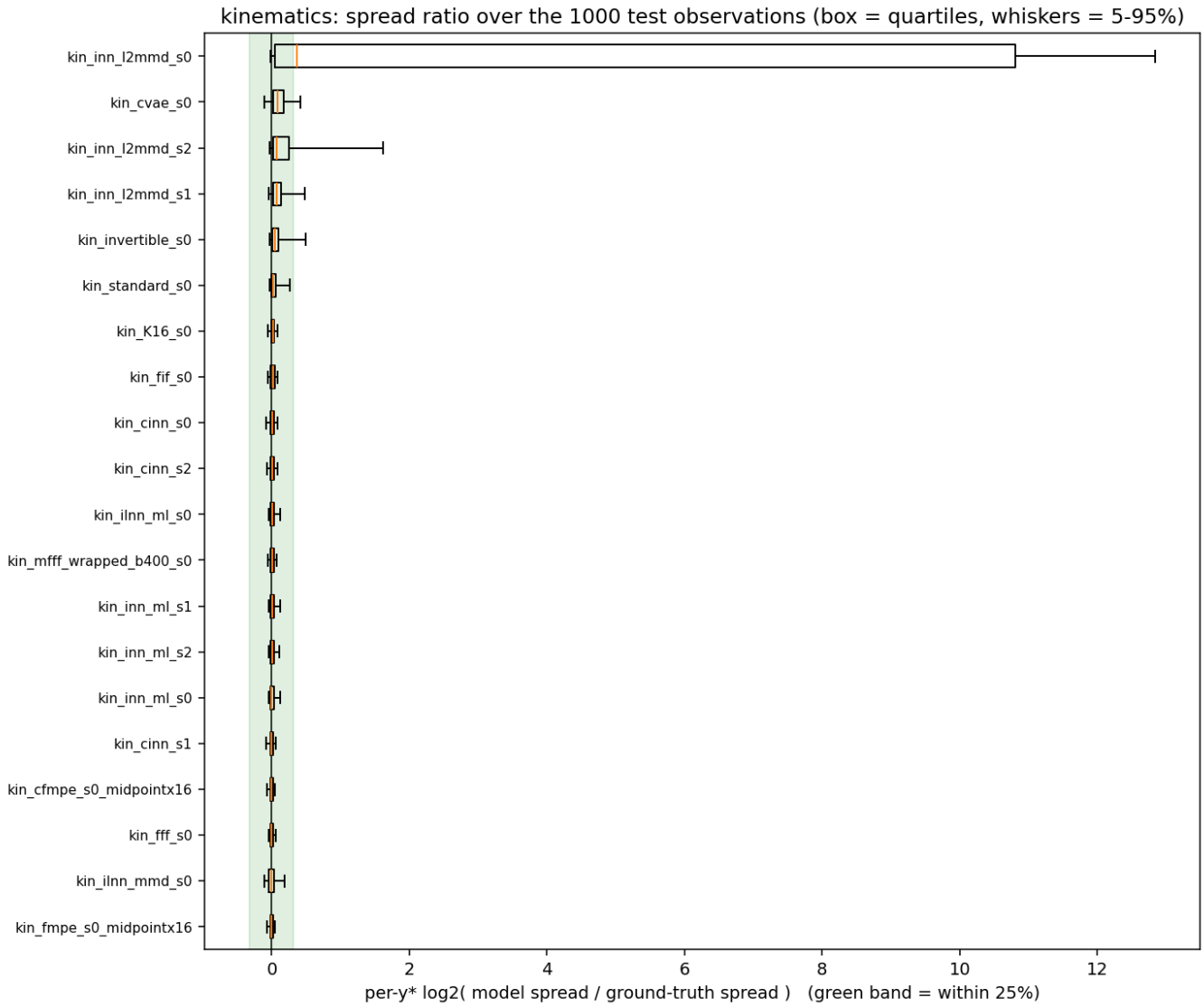
Kinematics 3: Signed coverage error per model



Bars show signed coverage error for marginal (blue) and joint (orange) coverage; the green band is the ±0.03 tolerance. Bars to the left are over-confident and bars to the right are under-confident.

Marginal errors (blue) are all inside the band. Joint errors (orange) tell a different story: **ilnn_mmd** (-0.115) and **invertible** (-0.074) are strongly over-confident, followed by **inn_l2mmd** and **inn_ml** seeds at about -0.044 to -0.064. **fff**, **fif** and **mfff_wrapped_b400** are slightly under-confident on the joint measure, at +0.031 to +0.036. The models with the smallest errors on both measures include the flow-matching variants **cfmpe** and **fmpe**, **cinn**, and **cvae**.

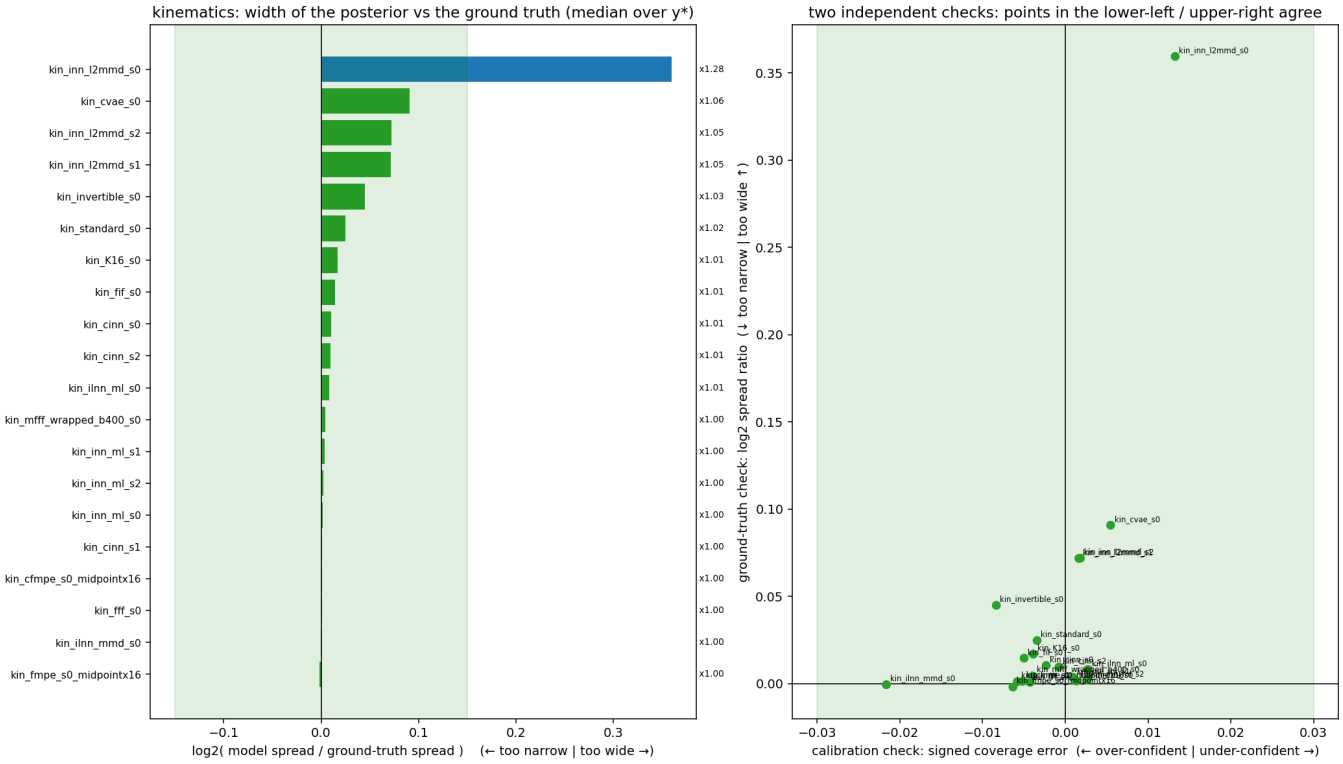
Kinematics 4: Spread ratio over the 1000 test observations



Distribution of $\log_2$(model spread / ground-truth spread) across test observations. The green band marks ±25%.

The x-axis is stretched by one outlier: **kin_inn_l2mmd_s0** has a box extending to about 10.8 on the $\log_2$ scale, meaning that for a large share of test observations its posterior is orders of magnitude wider than the truth. All other models are compressed near zero. **kin_inn_l2mmd_s2** and **kin_cvae_s0** have small positive medians with some right tail; the rest are tightly centred inside the band.

Kinematics 5: Posterior width vs ground truth, and agreement with coverage



Left: median $\log_2$ spread ratio per model. Only **kin_inn_l2mmd_s0** (×1.28) is outside the ±0.15 band (blue). **kin_cvae_s0** (×1.06) and the other l2mmd seeds (×1.05) are slightly wide but acceptable; the rest are at ×1.00 to ×1.03.

Right: the x-axis is marginal coverage error and the y-axis is spread ratio. Points are in the lower-left or upper-right if the checks agree. **kin_inn_l2mmd_s0** is far up the y-axis but inside the calibration band, so the checks disagree for that model. This plot uses marginal error only, so it does not show the joint over-confidence seen in Plots 1 and 3. **kin_ilnn_mmd_s0** has the largest marginal error (about -0.022) but a spread ratio of 1.00.

Key takeaways

1. Ballistics is mostly well calibrated. The one clear failure is the **inn_ml** family (all three seeds), which is under-confident on every check, with posteriors about 30% to 40% wider than the truth. **bal_cinn_s0** is borderline and worth watching.

2. Kinematics hides its problems in the joint view. Marginals are calibrated for all models, but ten models are over-confident jointly. If only marginal checks were reported, these would be missed.

3. Spread and coverage checks can disagree. The cases to inspect are **bal_cinn_s0** and **kin_inn_l2mmd_s0**, where the posterior is too wide even though the coverage looks calibrated.

4. The most consistent models across both benchmarks are the flow-matching variants (**cfmpe**, **fmpe**) and **cinn** seed 1, which are calibrated on all measures in both settings.

5. The sample size is small for some conclusions. Many models have a single seed (s0). Differences of a few hundredths in coverage error can be sampling noise, so seed-to-seed spread (for example across the inn_ml and l2mmd seeds) should be kept in mind.